

# Samuel James Hale

919-904-3651 | samueljameshale1998@gmail.com | LinkedIn | Portfolio | US Citizen

## Summary

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Full Stack Engineer experienced in building applications end-to-end, from interface to database. Delivered student-facing features as part of a 12-person team, resolved UI bugs, and built automation that ensured consistent outputs regardless of operator. Brings a practical mix of front-end polish, back-end logic, and deployment discipline to web products, along with hands-on experience integrating AI tooling — including LLM-powered features, prompt engineering, and AI-assisted development workflows — to accelerate delivery and improve product quality.

## Relevant Experience & Projects

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**AI Researcher & Programmer — ECU Faculty Research Group | (LLMs, NLP, Prompt Engineering, Smart Home AI)** 2025 – Present

[github.com/SamuelH98/Ultimate-SWE](https://github.com/SamuelH98/Ultimate-SWE)

- Built and open-sourced Ultimate SWE, a structured AI-agent workflow skill now adopted by the research group to guide coding agents through requirements, architecture, and testing in a disciplined order.
- Drive technical direction for the group by evaluating and integrating LLMs, NLP, and computer vision into smart home systems, turning promising tools into working prototypes.
- Trusted to independently research and prototype AI-powered features using cloud APIs, informing which approaches the group pursues further.

**Full-Stack Developer — Senior Capstone | (Django, PostgreSQL, Python, GitLab CI/CD)** Aug. 2024 – May 2025

NC Math Placement Web Application · Team of 12

- Delivered full-stack features — front end and back end — as part of a 12-person team building a web app to place NC students into math courses.
- Designed and built a document auto-generation feature that eliminated a manual, error-prone process, ensuring consistent output regardless of who ran it.
- Resolved user-facing UI bugs and helped maintain a CI/CD pipeline that caught issues automatically before they reached later stages.

## Technical Skills

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**Languages:** Python, JavaScript/TypeScript, Java, C#, C/C++, SQL, Go  
**Web & Backend:** React, Angular, Node.js, Django, Spring Boot, Flask, .NET, REST/GraphQL, microservices  
**Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Jenkins, GitLab, Azure DevOps)  
**Databases:** PostgreSQL, MySQL, SQL Server, Redis, Neo4j  
**AI & Tooling:** Claude Code, GitHub Copilot, LLM-assisted development, prompt engineering  
**Practices:** Agile/Scrum, TDD, secure coding (OWASP, OAuth 2.0/JWT), Git/GitHub

## Education

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**East Carolina University — M.S. in Computer Science** Jan. 2023 – May 2026

**East Carolina University — B.S. in Software Engineering, Concentration in Data Science** Aug. 2021 – May 2025

Chancellor's List (Spring 2025), Dean's List (Fall 2022, Fall 2023) · BS/MS Accelerated Program